

Labeling Tutorial

Resources Provided:

You will receive the following resources:

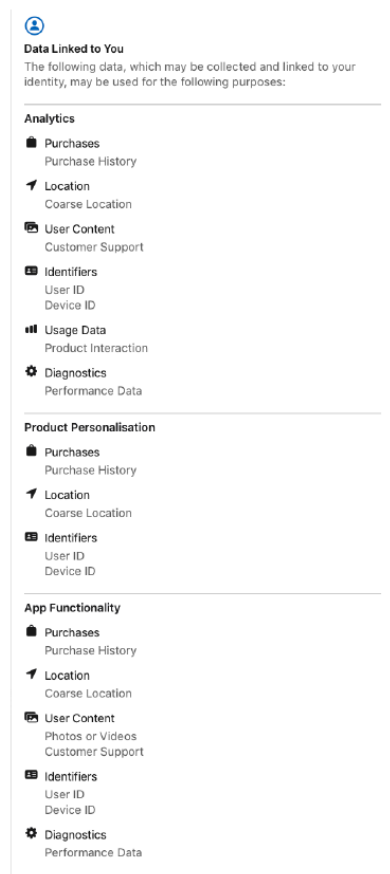
For each app, we will give you one kind of privacy information (refer to the .csv column ['pp_category']). Currently, we recognize seven types of privacy—'CONTACT', 'CAMERA', 'MICROPHONE', 'STORAGE', 'CALENDAR', 'SMS', 'LOCATION'. In addition, we have crawled numerous information resources (e.g., app description, app summary, privacy policy, etc.) from the web.

Objective:

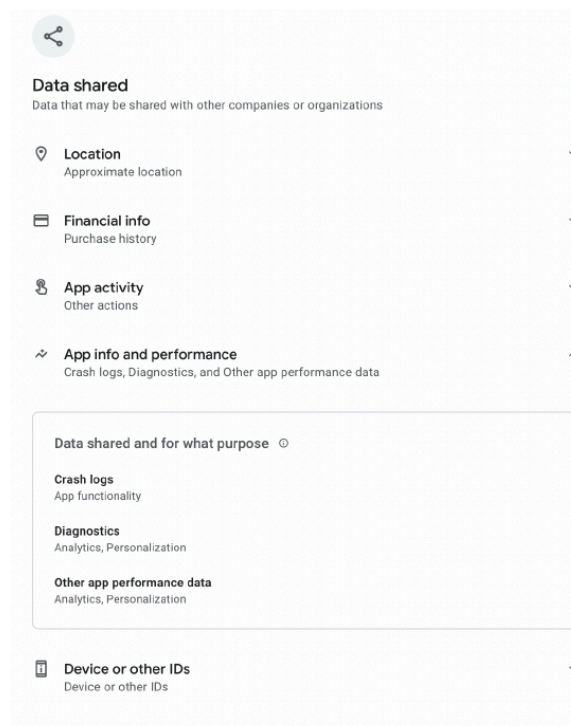
Our goal is to create an industry-standard dataset. Each ground truth within it should be articulated in natural language (which is where your expertise comes into play). These entries will detail the specific uses, methods, and functions associated with an app's handling of privacy data.

Distinguishing Features from Existing Data:

Both Google Play and Apple Store feature sections dedicated to app privacy practices. In Google Play, this part is called **Data Safety**. However, they merely offer a broad overview of the privacy information usage, lacking depth and specificity.



(a) iOS App Privacy Label



(b) Android Data Safety Label

The dataset we aim to create offers significantly richer information. Moreover, the characteristics of natural language are well-suited to meet the current demands and training needs of large language models.

To produce a high-quality dataset, especially a reliable ground truth, we require top-tier data. In the initial stage where manual annotation is critical, our information primarily comes from app stores like Google Play and official app websites. Annotations during this stage mainly rely on the description & summary from the Google Play page and the privacy policy from app websites. You will also need to consider information from Google Play's data safety section.

App summary & description

The App summary & description generally contain:

1. Summary: This typically provides a brief one-liner that offers an overview of the app's main functionality or selling point.
2. Description: This gives users an in-depth understanding of the application, helping them grasp its full scope. The description usually includes:
 - a. Feature List: A detailed breakdown of the primary features and functionalities of the app. For instance, a photo editing tool might list "auto-beautification, filter effects, cropping, and rotation" among its features.
 - b. Use Cases: This depicts scenarios or contexts in which a user might find the app beneficial. It usually includes examples of such scenarios.
 - c. Technical Specifications: The technical implementation behind the application's features., such as "low-power background processing."

While these descriptions usually don't directly address the app's privacy practices or detailed information about user data collection, sharing, and utilization, the listed functionalities and use-cases often implicitly reveal specific purposes related to privacy. The technical details mentioned may hint at how private information is precisely used.

For instance, if an app requires access to a user's location to offer a particular service, developers might showcase features related to user location in the description. A feature like 'Share your terrain images and current location with the community' subtly indicates the purpose of using location privacy. Meanwhile, 'View your current latitude, longitude, and altitude at any moment' suggests the specific method of using location data, such as through GPS.

Privacy Policy

A well-written Privacy Policy generally includes: (The bolded points are the main content we are concerned with)

1. **Information Collection and Usage:**

- What types of personal information are collected (e.g., name, email address, location data, etc.).

- How and why these pieces of information are collected, including information collection through cookies and other tracking technologies.

2. ****Privacy Practices for Specific Features and Services****:

- If the app offers specific features and services, like geolocation services, describe the specific privacy practices for these features and services.

3. ****Information Sharing and Disclosure****:

- Situations where user information is shared or disclosed with third parties (e.g., advertisers and analytics service providers).
- The purposes for which the information might be used, like targeted advertising or analytics.

4. ****User Permissions and Choices****:

- How users can access, correct, update, or delete their personal information.
- How users can opt out of certain information collection or usage, like opting out of receiving marketing communications.

5. ****Third-Party Services and Content****:

- How third-party services, like analytics and advertising services, use user information.
- How third-party websites or services can access user data through the application.

6. ****Data Protection and Security****:

7. ****International Data Transfers****:

8. ****Children's Privacy****:

9. ****Changes to the Privacy Policy****:

10. ****Contact Information****:

11. ****Legal and Regulatory Compliance****:

12. ****Definitions and Explanations****:

Compared to the App summary & description, the privacy policy typically doesn't delve into detailed functionalities. However, it provides a clear statement of data collection, sharing, and the contexts in which it's applied.

Details and Requirements for the Annotation Task:

Below are the details and specific steps for the annotation task we have provided you.

1. Given one privacy information class. Choose key sentences from 1) summary+description (SD), 2) privacy policy (PP).
 - o For SD, we want to focus more on **Detailed Information Usage and Privacy Practices for Specific Features and Services**.

- For PP, we balance our focus on four parts: **1) Information Usage and Privacy Practices for Specific Features and Services. 2) Information Sharing and Disclosure 3) User Permissions and Choices 4) Third-Party Services and Content.**
- A key sentence refers to a sentence or paragraph that directly addresses (or implies/hints at) the content highlighted in bold above.
- 2. Rank the chosen sentences, from most important to less important. In list format ([‘sentence 1’, ‘sentence 2’, ..., ‘sentence n’], see demo.csv).
 - Important key sentences are characterized by their high **informational richness** and strong **relevance to the privacy information**.
- 3. Annotate keywords from des and pp, use two separate columns to write them. Also in list format. ([‘keyword 1’, ‘keyword 2’, ..., ‘keyword n’])
 - For keywords, it means 1) vocabulary of specific **actions** related to the privacy info that we are interested in (eg. sharing, store, collect, etc.), 2) **meaningful nouns**, such as specific privacy information (eg. For ‘CAMERA’ privacy, these nouns can be ‘photo’, ‘snapshot’, ‘video’, ‘album’, etc.), 3) vocabulary related to the specific **functions/features** of the app.
- 4. Summarize the information (from the key sentences of description & summary and privacy policy) in two versions.
 - **Detailed privacy info usage**
 - i. If the app has multiple usage scenarios regarding privacy information, use more than one point.
 - ii. Try to cover as many keywords as possible.
 - **Concise version**
 - i. Summarize in one sentence how the app uses users' privacy information.
 - For the detailed summary:
 - i. Requirement for content: it should primarily cover Information Usage and Privacy Practices. Additionally, if provided in the resources, it should touch upon Information Sharing and Disclosure, User Permissions and Choices, and Third-Party Services and Content.
 - ii. Other requirements:
 - 1. Aim for a high semantic coverage of the keywords. Avoid including unnecessary details and numbers (eg. specs).
 - 2. Adhere strictly to the key sentences/keywords and avoid adding uncertain information outside the scope of the provided material.
 - iii. Format requirement: use list format. [‘point 1’, ‘point 2’, ..., ‘point n’]
 - For the concise summary:
 - i. Requirement for content: Provide one sentence that summarizes the app's Information Usage and Privacy Practices.
 - ii. Format requirement: also use list format.
- 5. Please rate the quality of this app's description and privacy policy on a scale of 0-5. Additionally, provide comments to clarify the reason for your rating.
 - 0: No privacy policy. The description offers little or no useful information.

- 1-2: The privacy policy is either very short or lacks substantial information. The description provides minimal information and hardly any specific privacy use cases or feature descriptions.
- 3-4: The privacy policy adequately summarizes how privacy is handled. The description includes specific privacy use cases and detailed feature introductions.
- 5: The privacy policy is well-organized, provides more than just information about privacy practices; it offers other beneficial insights. The description is comprehensive, with numerous specific privacy use cases and feature details.